!

Lecture 7? - DS

24.04.2026

I. Linear homogeneous differential systems with constant coeff.

- gen. solution
- type, stability

(1): $x' = Ax$, where $A \in \mathcal{M}_m(\mathbb{R})$

Theorem 1: For each vector $\eta \in \mathbb{R}^m$, the IVP $x' = Ax$, $x(0) = \eta$ has a unique solution, $\varphi : \mathbb{R} \to \mathbb{R}^m$ given by $\boxed{\varphi(t) = e^{tA} \cdot \eta}$

! **Remark**: The general sol. of $x' = \underline{A}x$ can be ... $= c$.
$c \in \mathbb{R}^m$ arbitrary.

- $x' = Ax$, $x(0) = \eta \Rightarrow$
  $\Rightarrow \varphi(t) = e^{tA} \cdot \eta$ (unique)
- e^{tA} when A diagonalizable
- $x' = f(x)$; regular point; $f(\eta) \neq 0$,
  ↳ sol: $\varphi : I \to \mathbb{R}$, $\varphi'(t) = f(\varphi(t))$
- $\varphi'(t) = f(\varphi(t))$
- IVP → unique sol.
- η^* equil. of $x' = f(x) \Rightarrow$
  $\Rightarrow$ unique sol: $\varphi(t) = \eta$ where $\begin{cases} x' = f(x) \\ x(t_0) = \eta \end{cases}$

Recall how we find $e^{t \cdot A}$ when A is diag...

→ Step 1: Find eigenvalues, $\lambda_1, \dots, \lambda_m \in \mathbb{C}$ a... ...ng eigenvectors $u_1, \dots, u_m \in \mathbb{C}^m$

→ Step 2: Decide whether $u_1, \dots, u_m$ are lin. independent:

$P = (u_1, \dots, u_m)$, compute $\det(P)$
(matrix)

If $\det(P) \neq 0 \Rightarrow \{u_1, \dots, u_m\}$ lin. indep. $\Rightarrow$ A diagonalizable over $\mathbb{C}$.

Assume $\det P \neq 0$.

→ Step 3: $P^{-1} = ?$

Consider $D = \operatorname{diag}(\lambda_1, \dots, \lambda_m)$

Write $e^{tD} = (e^{t\lambda_1}, \dots, e^{t\lambda_m})$

→ Step 4: $e^{tA} = P \cdot e^{tD} \cdot P^{-1}$, $(\forall) t \in \mathbb{R}$

Proposition: Assume that A - diagonalizable over $\mathbb{R}$ and let $u_1, \dots, u_m \in \mathbb{R}^m$ be the m lin. indep. eigenvectors corresponding to the eigenvalues $\lambda_1, \dots, \lambda_m \in \mathbb{R}$. Then, the gen. sol. of the system $x' = Ax$ is:

$$x = c_1 \cdot e^{t \cdot \lambda_1} \cdot u_1 + \dots + c_m \cdot e^{t \cdot \lambda_m} \cdot u_m \ ; \quad c_1, \dots, c_m \in \mathbb{R}.$$

In particular, $e^{t\lambda_1} u_1, \dots, e^{t\lambda_m} u_m$ are m lin. indep. sol. of $x' = Ax$

Proof.

Let $x(t)$ be a sol. of $x' = Ax$

Let $\eta = x(0)$

Then $x(t) = e^{tA} \cdot \eta$ (1)

- Let $P = (u_1, \dots, u_m) \in \mathcal{M}_m(\mathbb{R})$

  $D = \operatorname{diag}(\lambda_1, \dots, \lambda_m) \in \mathcal{M}_m(\mathbb{R})$

$\hookrightarrow$ $e^{tA} = P \cdot e^{tD} \cdot P^{-1}$ (2)

(2) $\to$ (1): $x(t) = \underbrace{P \cdot e^{tD} \cdot P^{-1}} \cdot \eta = P \cdot e^{tD} \cdot c$, where $\overbrace{P^{-1} \cdot \eta}^{c}$

Notation: $c = P^{-1} \cdot \eta = \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix}$

$$x(t) = \underbrace{(u_1 \dots u_m)}_{P} \underbrace{\begin{pmatrix} e^{t\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{t\lambda_m} \end{pmatrix}}_{e^{tD}} \underbrace{\begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix}}_{c = P^{-1} \cdot \eta}$$

$$x(t) = \left(e^{t\lambda_1} \cdot u_1 \dots e^{t\lambda_m} \cdot u_m \right) \cdot \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix}$$

$$x(t) = c_1 \cdot e^{t\lambda_1} \cdot u_1 + \dots + c_m \cdot e^{t\lambda_m} \cdot u_m$$

Take $c_1 = 1$, $c_2 = c_3 = \dots = c_m = 0 \Rightarrow e^{t\lambda_1} \cdot u_1$ is a solution.

Similarly, $e^{t\lambda_m} \cdot u_m$ is a solution (all the rest)

- We prove now that $e^{t\lambda_1} \cdot u_1, \dots, e^{t\lambda_m} \cdot u_m$ are lin. indep. functions from $\underbrace{C(\mathbb{R}, \mathbb{R}^m)}_{\text{the linear space}}$. We use the definition.

$\hookrightarrow$ Take $c_1, \dots, c_m \in \mathbb{R}$ st. $c_1 \cdot e^{t\lambda_1} \cdot u_1 + \dots + c_m \cdot e^{t\lambda_m} u_m = 0$, $(\forall) t \in \mathbb{R}$

$\xRightarrow{t=0}$ $c_1 \cdot u_1 + \dots + c_m \cdot u_m = 0$

$\Rightarrow c_1 = c_2 = \dots = c_m = 0 \Rightarrow u_1, \dots, u_m$ lin. indep. on $\mathbb{R}^m$

Theorem 2. Let $A \in M_m(\mathbb{R})$. Assume that $Re(\lambda) < 0$, $(\forall)\ \lambda$ eigenvalue of A.

Then $\lim_{t\to\infty} e^{tA} = O_3$, hence $\lim_{t\to\infty} x(t) = 0 \in \mathbb{R}^m$ for any solution $x(t)$ of $x' = Ax$

Proof: (for A diag. over $\mathbb{C}$).

- we consider just the case that A is diag. over $\mathbb{C}$.
- we use the notations presented before.

$$e^{tA} = \underset{ct.}{P}\, e^{tD}\, \underset{ct.}{P^{-1}}, \quad (\forall)t \in \mathbb{R}.$$

$\Rightarrow \lim_{t\to\infty} e^{tA} = P.O_3.P^{-1} = O_3$

$x(t) = e^{tA}.x(0)$ $\Rightarrow$

THE NULL MATRIX

We claim $\lim_{t\to\infty} e^{tD} = O_3$

$\Rightarrow \lim_{t\to\infty} x(t) = 0 \in \mathbb{R}^3$ for any solution.

- $\boxed{\text{Let } \lambda \in \mathbb{C}}$ with $Re(\lambda) < 0$. $\lim_{t\to\infty} e^{t\lambda} = ?$

$\lambda = \underset{Re(\lambda)}{\alpha} + i\beta$ with $\alpha, \beta \in \mathbb{R}$, $\underline{\alpha < 0}$.

$$e^{t\lambda} = e^{t(\alpha+i\beta)} = e^{t\alpha + i\beta t} = e^{\alpha t}.(\cos\beta t + i\sin\beta t) = e^{\alpha t}.\cos\beta t + i e^{\alpha t}.\sin\beta t.$$

$-1 \le \cos\beta t \le 1$, $(\forall)t \in \mathbb{R}$ $|\cdot e^{\alpha t}$

$-e^{\alpha t} \le e^{\alpha t}.\cos\beta t \le e^{\alpha t}$, $(\forall)t \in \mathbb{R}$

$\alpha < 0$

$\underline{\alpha < 0} \Rightarrow \lim_{t\to\infty} e^{\alpha t} = 0$

0

$\Rightarrow \lim_{t\to\infty} e^{\alpha t}.\cos\beta t = \lim_{t\to\infty} e^{\alpha t}.\sin\beta t = 0 \Rightarrow \lim_{t\to\infty} e^{t\lambda} = 0$

$\Rightarrow$ Then $\lim_{t\to\infty} e^{tD} = O_3$.

The qualitative study of first order scalar autonomous d.e.

$\boxed{(1) \quad x' = f(x),}$ where $f: \mathbb{R} \to \mathbb{R}$ is a C^1 function.

Example: $x' = 1 - x^2$. We have $f(x) = 1 - x^2$, $f: \mathbb{R} \to \mathbb{R}$
(non-linear)

Def: An equilibrium point of (1) is $\eta^* \in \mathbb{R}$ s.t. $f(\eta^*) = 0$.

A point $\eta^* \in \mathbb{R}$ which is not equilibrium is called a regular point.

Example: -1 and 1 are equilibrium points for $\dot{x} = 1 - x^2$.

Def: A solution for d.e. (1) is a function $\varphi: I \to \mathbb{R}$ s.t. $I \subset \mathbb{R}$ is an open interval, φ is C^1 and $\varphi'(t) = f(\varphi(t))$, $(\forall) t \in I$

Theorem 1. Let $t_0 \in \mathbb{R}$ and $\eta \in \mathbb{R}$. The IVP $\begin{cases} x' = f(x) \\ x(t_0) = \eta \end{cases}$ has a unique solution, $\varphi: (\alpha, \beta) \to \mathbb{R}$ defined on a maximal interval (α, β).

- Moreover, when φ is bounded in the future ($\varphi|_{[t_0, \beta)}$ is bounded), we have that $\beta = +\infty$. (t_0: initial time)
- Similarly, when φ is bounded in the past, ($\varphi|_{(\alpha, t_0]}$ is bounded), then $\alpha = -\infty$.

Property 1: Let $\eta^* \in \mathbb{R}$ be an equil. point of (1). Then the unique sol. of IVP $\begin{cases} x' = f(x) \\ x(t_0) = \eta \end{cases}$ is $\varphi: \mathbb{R} \to \mathbb{R}$, $\varphi(t) = \eta$, $(\forall) t \in \mathbb{R}$.

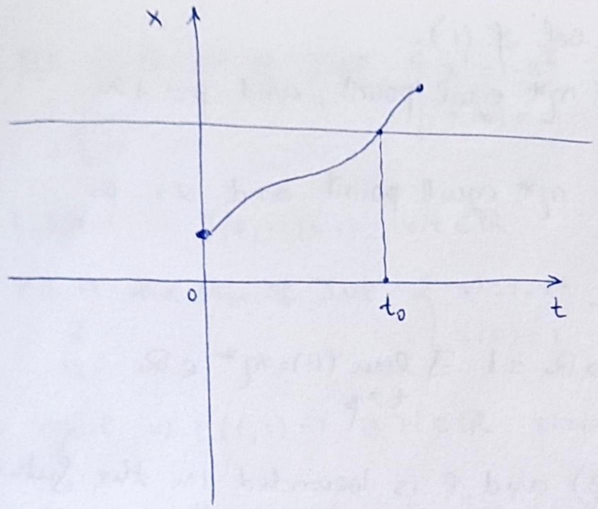
WHY:

$\left(\begin{array}{l} f(\eta^*) \overset{def}{=} 0 \\ \text{deriv. of } (x') = 0 \end{array} \right)$

Property 2: Let $\eta^* \in \mathbb{R}$ be an equil. point and $\varphi: I \to \mathbb{R}$ be a sol. of (1) st. $0 \in I$ and $\varphi(0) < \eta^*$. Then $\varphi(t) < \eta^*$, $(\forall) t \in I$.

$\varphi(0) > \eta^* \qquad \varphi(t) > \eta^*$

Proof:



($\exists!$ → graficele a două soluții $\neq$ nu se pot intersecta)

Assume, by contradiction, $\exists t_0 \in I$ s.t. $\varphi(t_0) \geq \eta^*$

$\Rightarrow$ since φ is cont., $\exists t_0 \in I$ s.t. $\varphi(t_0) = \eta^*$

Consider IVP:

$$\begin{cases} x' = f(x) \\ x(t_0) = \eta^* \end{cases}$$

We have φ is a sol. of this IVP.
the constant function η^* is also a sol. $\Big\} \xrightarrow{\text{th 1}}$ $\varphi(t) = \eta^*$, $(\forall) t \in \mathbb{R}$ ← uniqueness

but it contradicts the hyp. $\varphi(0) < \eta^*$.

Prop3: Let $\varphi : I \to \mathbb{R}$ be a sol. of (1) with $0 \in I$ and $\varphi(0)$ a regular point (not equil.).
Then $\varphi'(t) = f(\varphi(t)) \neq 0$, $(\forall) t \in I$, hence φ is strictly monotone. $\Rightarrow$ (no extremum points) (t0)

Proof.

$\varphi(0)$ regular point. $\Rightarrow f(\varphi(0)) \neq 0$.

Assume by contradiction that $\exists t_0 \in I$ s.t. $f(\varphi(t_0)) = 0$

$\Rightarrow \varphi(t_0)$ equil. point.

Denote $\eta^* = \varphi(t_0)$ equil. point.

Consider IVP $\begin{cases} x' = f(x) \\ x(t_0) = \eta^* \end{cases}$ which has 2 solutions: φ ; ct. fct. η^*

$\Rightarrow \varphi(t) = \eta^*$, $(\forall) t \in \mathbb{R}$, this contradicts $\varphi(0)$ regular point.

Prop 4: Let $\varphi : (\alpha, \beta) \to \mathbb{R}$ be a sol. of (1).

If $\exists \lim_{t \nearrow \beta} \varphi(t) = \eta^* \in \mathbb{R}$ then η^* equil. point, and $\beta = +\infty$

If $\exists \lim_{t \searrow \alpha} \varphi(t) = \eta^* \in \mathbb{R}$ then η^* equil. point and $\alpha = -\infty$

Proof.

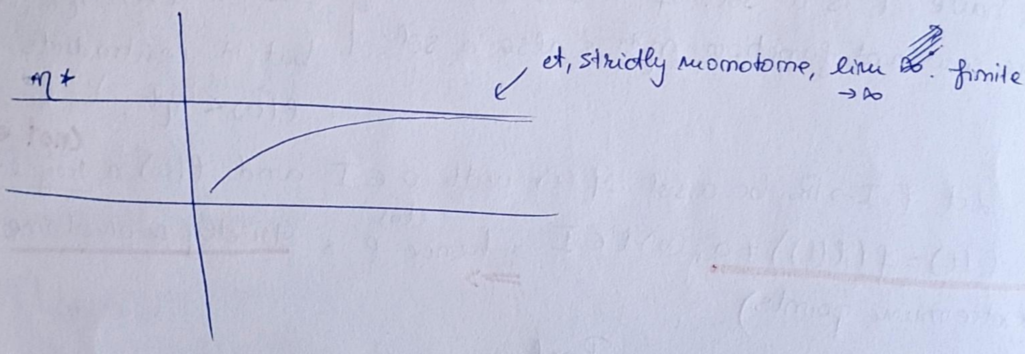
We have a solution $\varphi : (\alpha, \beta) \to \mathbb{R}$ s.t. $\exists \lim_{t \nearrow \beta} \varphi(t) = \eta^* \in \mathbb{R}$.

$\hookrightarrow \varphi'(t) = f(\varphi(t)),\ (\forall) t \in (\alpha, \beta)$ and φ is bounded in the future.

$\overset{\text{th 1}}{\Longrightarrow} \beta = +\infty$

$$\left.\begin{array}{l} \varphi : (\alpha, \beta) \to \mathbb{R},\quad \varphi'(t) = f(\varphi(t)),\ (\forall) t \in (\alpha, \beta) \\ \lim_{t \to \infty} \varphi(t) = \eta^* \overset{f \text{ cont.}}{\Longrightarrow} \lim_{t \to \infty} f(\varphi(t)) = f(\eta^*) \end{array}\right\} \Rightarrow$$

$\Rightarrow \lim_{t \to \infty} \varphi'(t) = f(\eta^*)$



We claim $\lim_{t \to \infty} \varphi'(t) = 0$ in these hypotheses.

$\Rightarrow f(\eta^*) = 0 \Rightarrow \eta^*$ equil.

Lemma: Let $\varphi : (\alpha, +\infty) \to \mathbb{R}$ be a C^1 function s.t. $\exists \lim_{t \to \infty} \varphi(t) \in \mathbb{R}$ and $\lim_{t \to \infty} \varphi'(t) \in \mathbb{R}$. Then $\lim_{t \to \infty} \varphi'(t) = 0$. **(Barbălat)**

(next time)

Example: $x' = 1 - x^2$. Denote $\varphi(\cdot, \eta)$ the sol. of the IVP

$$\begin{cases} x' = 1 - x^2 \\ x(0) = \eta \end{cases}$$

a) find $\varphi(t, -1)$ and $\varphi(t, 1)$

b) prove $-1 < \varphi(\cdot, 0) < 1$, $(\forall) t \in \mathbb{R}$

$\varphi(\cdot, 0)$ strictly increasing, $\lim_{t \to \infty} \varphi(t, 0) = 1$ and $\lim_{t \to -\infty} \varphi(t, 0) = -1$

We found the equil. points of $x' = 1 - x^2$: $1, -1$

a) $\varphi(t,-1)$ the sol. of IVP $\begin{cases} x' = 1 - x^2 \\ x(0) = -1 \end{cases}$, $\ast$

$\downarrow$

η

$\eta = -1$ equil. $\Rightarrow \varphi(t,-1) = -1$, $(\forall) t \in \mathbb{R}$.

$\varphi(t,1)$ the sol. of IVP $\begin{cases} x' = 1 - x^2 \\ x(0) = 1 \end{cases}$

$\downarrow$

η

$\eta = 1$ equil. $\Rightarrow \varphi(t,1) = 1$, $(\forall) t \in \mathbb{R}$ since 1 is equil.

b) $\varphi(t,0)$ is the sol. of IVP: $\begin{cases} x' = 1 - x^2 \\ x(0) = 0 \end{cases}$

$\parallel$

$\varphi(t)$

$\varphi(t) = \varphi(t,0)$, $t \in I$.

$\begin{cases} \varphi'(t) = f(\varphi(t)), \ (\forall) t \in I \\ \varphi(0) = 0 \end{cases}$ $\rightarrow$ regular point $\Rightarrow$ not a c. fct. ??!

$-1 < \varphi(0) < 1$ (obvs) $\xRightarrow{\text{Prop 2}}$ $-1 < \varphi(t) < 1$, $(\forall) t \in I$.

$\Rightarrow \varphi$ is bounded $\xRightarrow{\text{th. 1}}$ $I = \mathbb{R}$.

$\Rightarrow \varphi : \underset{=}{I} \rightarrow \mathbb{R}$

φ is <u>not</u> a cont. fct $\Rightarrow \varphi$ is strictly monotone. $\Bigg\}$ $\Rightarrow \varphi$ is strictly increasing

$\Rightarrow f(\varphi(t)) > 0$, $(\forall) t \in \mathbb{R}$